

Problem 1:

$$\text{已知} \begin{cases} \sum_{j=1}^m p_{ij} = r_i & \forall i=1, \dots, n \end{cases}$$

$$\begin{cases} \sum_{i=1}^n p_{ij} = c_j & \forall j=1, \dots, m \end{cases}$$

有 $m+n$ 个方程

$$\text{最优化 } \sum_{i=1}^n \sum_{j=1}^m p_{ij} c_{ij}$$

标准线性规划形式如下

$$\text{已知 } AX=b, \text{ 最优化 } C^T X$$

$$\text{即可令 } \bar{X}' = (p_{11}, p_{12}, \dots, p_{1m}, p_{21}, p_{22}, \dots, p_{2m}, \\ p_{n1}, p_{n2}, \dots, p_{nm}) \quad \bar{X}' \in R^{mn}$$

$$\bar{C}' = (c_{11}, c_{12}, \dots, c_{1m}, c_{21}, \dots, c_{2m}, \\ c_{n1}, \dots, c_{nm}) \in R^{mn}$$

$$\bar{b}' = (r_1, \dots, r_n, c_1, \dots, c_m) \in R^{m+n}$$

$$A = R^{(m+n) \times mn}$$

$$\text{前 } n \text{ 行: } A_{ik} = \begin{cases} 1 & \text{if } k=(i-1)m+j, j=1,2,\dots,m \\ 0 & \text{其他} \end{cases}$$

(行约束)

$$\text{后 } m \text{ 行: } A_{(n+j),k} = \begin{cases} 1 & \text{if } k=(i-1)m+j, i=1,2,\dots,m \\ 0 & \text{其他} \end{cases}$$

(列约束)

$$\text{对偶问题为 } A^T y \in C \rightarrow \max b^T y$$

$$\bar{y} \in R^{n+m} \text{ 为对偶何量}$$

优化

设 $\vec{y} = (f_1, f_2, \dots, f_n, g_1, \dots, g_m)$

$$\text{有 } b^T y = \sum_{i=1}^n r_i f_i + \sum_{j=1}^m c_j g_j$$

$$A^T y = c \text{ 对应 } \underbrace{f_i + g_j \leq c_{ij}}_{\substack{i=1, \dots, n \\ j=1, \dots, m}}$$

对偶问题即为 已知 $\begin{cases} f_i + g_j \leq c_{ij} \\ f_i \in \mathbb{R}, g_j \in \mathbb{R} \end{cases} \quad \begin{matrix} i=1, \dots, n \\ j=1, \dots, m \end{matrix}$

$$\max \sum_{i=1}^n r_i f_i + \sum_{j=1}^m c_j g_j$$

Problem 2:

先证明 存在 $2n$ 个点构成的集合可以被打破

构造大小为 $2d$ 的点集

$$S = \{p_1^-, p_1^+, p_2^-, p_2^+, \dots, p_d^-, p_d^+\}$$

p_i^- 在第 i 个坐标上为 -1 , 其他为 0

p_i^+ 在第 i 个坐标上为 1 , 其他为 0

构造矩形 a, b 如下:

$$\text{取 } \begin{cases} L_i = \min(p_i \mid p \in Y) \\ U_i = \max(p_i \mid p \in Y) \end{cases}$$

$$\text{取 } \varepsilon < 1 \quad a_i = L_i - \varepsilon \quad b_i = U_i + \varepsilon$$

对 $\forall p \in \gamma$ 有 $a_i = L_i - \varepsilon \leq L_i \leq p_i$

$$b_i = U_i + \varepsilon \geq U_i \geq p_i$$

$a_i \leq p_i \leq b_i$ 对所有 i 成立

对 $p \in \gamma$ ，由于每个 p 均存在在某个维度上的
最大值或最小值，且比第二小的大 1 以上

在该维度 k ，必有 $p_k < a_j$ 或 $p_k > b_j$

故该大小为 $2d$ 点集可被打散

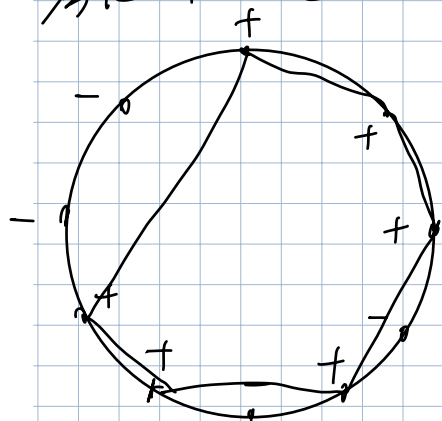
再证明大小为 $2d+1$ 点集不可被打散：

对任意的 $2d+1$ 点集，取 d 个维度最大和最小值共 $2d$
个点，其必然包括剩下的 1 个点，故无法被打散

综上所述， H 的 VC dimension 为 $2d$

Problem 3

考虑单位圆上的 K 个点构成的集合 S



对任意 $A \subset S$

以其所有点为顶点的凸多边形

将 A 与 $S \setminus A$ 分开

$K \rightarrow +\infty$ 时仍成立

极平面上的凸多边形 VC dimension 是无穷

Problem 4

1. 即为蓄水池抽样, 算法如下:

① 在时刻 1, 维护 $\hat{a} = a_1$ 并输出

② 在时刻 t , 以概率 $\frac{1}{t}$ 将 $\hat{a}$ 更新为 a_t
否则 $\hat{a}$ 的值不变

需要维护 $\hat{a}$ 和 t , 故需要 $O(\log n + \log m)$ bit 空间

2. 当 $t=1$ 时, 以概率 1 输出 a_1

假设当 $t=t_0$ 时 输出的值为 $\hat{a}_{t_0}$

$$\text{满足 } P(\hat{a}_{t_0} = a_i) = \frac{a_i^2}{\sum_{i=1}^{t_0} a_i^2}$$

则当 $t=t_0+1$ 时,

$$P(\hat{a}_{t_0+1} = a_i) = P(\hat{a}_{t_0+1} = \hat{a}_{t_0}) \cdot P(\hat{a}_{t_0} = a_i)$$

$$= (1 - P) \cdot \frac{a_i^2}{\sum_{i=1}^{t_0} a_i^2} = \frac{a_i^2}{\sum_{i=1}^{t_0+1} a_i^2}$$

$$\Rightarrow P = 1 - \frac{\sum_{i=1}^{t_0} a_i^2}{\sum_{i=1}^{t_0+1} a_i^2} = \frac{a_{t_0+1}^2}{\sum_{i=1}^{t_0+1} a_i^2}$$

即将1中算法修正为

①在时刻1. 维护 $\hat{a} = a_1$ 并输出

②在时刻 t , 以概率 $\frac{a_t^2}{\sum_{i=1}^t a_i^2}$ 将 $\hat{a}$ 更新为 a_t
 否则 $\hat{a}$ 的值不变

Problem 5

$$\text{有 } \sum_{p \in S_j^t} \frac{|N_j^t|}{x} g(p, c) - \sum_{p \in N_j^t} g(p, c) \leq \varepsilon_0 2^{t+1} R |N_j^t|$$

两边求和有

$$\sum_{j=1}^m \sum_{t=0}^{\Phi} \left| \sum_{p \in S_j^t} \frac{|N_j^t|}{x} g(p, c) - \sum_{p \in N_j^t} g(p, c) \right|$$

$$\leq \sum_{j=1}^m \sum_{t=0}^{\Phi} \varepsilon_0 2^{t+1} R |N_j^t|$$

$$= \sum_{j=1}^m \sum_{t=1}^{\Phi} \varepsilon_0 2^{t+1} R |N_j^t| + \sum_{j=1}^m \varepsilon_0 2 R |N_j^0|$$

$$\leq 4 \varepsilon_0 n f(p, A) + \sum_{j=1}^m \varepsilon_0 2 R |N_j^0|$$

$$\text{即证 } \sum_{j=1}^m \varepsilon_0 2 R |N_j^0| = \Theta(\varepsilon_0) n f(p, A)$$

$$\sum_{j=1}^m |N_j^0| = \sum_{j=1}^m \left\lceil \log \frac{d(p, a_i)}{R} \right\rceil \sim n f(p, A)$$

$$\text{即原式} = \varepsilon_0 \cdot 2R \underbrace{A}_{\text{面积}} n f(p, A) = O(\varepsilon_0) n f(p, A)$$

得证.

Problem 6 $|C|=K$ 为 K 个中心点

$$\text{有 } (1-\varepsilon) f(S_1, C) \leq f(A_1, C) \leq (1+\varepsilon) f(S_1, C)$$

$$(1-\varepsilon) f(S_2, C) \leq f(A_2, C) \leq (1+\varepsilon) f(S_2, C)$$

$$\text{由定义 } f(A_1, C) = \sum_{S \in S_1} w_1(S) \min_{C \in C} \|S - C\|^2$$

$$f(A_2, C) = \sum_{S \in S_2} w_2(S) \min_{C \in C} \|S - C\|^2$$

$$f(A_1 \cup A_2, C) = \sum_{S \in S_1 \cup S_2} w_{12}(S) \min_{C \in C} \|S - C\|^2$$

$$= \sum_{S \in S_1 \setminus S_2} w_{12}(S) \min_{C \in C} \|S - C\|^2$$

$$+ \sum_{S \in S_2 \setminus S_1} w_{12}(S) \min_{C \in C} \|S - C\|^2$$

$$+ \sum_{S \in S_1 \cap S_2} w_{12}(S) \min_{C \in C} \|S - C\|^2$$

$$\text{由题知 } w_{12}(S) = \begin{cases} w_1(S) & X \in S_1 \setminus S_2 \\ w_2(S) & X \in S_2 \setminus S_1 \\ w_1(S) + w_2(S) & X \in S_1 \cap S_2 \end{cases}$$

$$= \sum_{S \in S_1 \setminus S_2} w_1(S) \min_{C \in C} \|S - C\|^2$$

$$+ \sum_{S \in S_1 \cap S_2} (w_1(S) + w_2(S)) \min_{C \in C} \|S - C\|^2$$

$$+ \sum_{S \in S_2 \setminus S_1} w_2(S) \min_{C \in C} \|S - C\|^2$$

同理

$$= f(A_1, C) + f(A_2, C) \quad \frac{f(S_1 \cup S_2, C)}{= f(S_1, C) + f(S_2, C)}$$

$$\Rightarrow (1-\varepsilon)f(S_1, C) \leq f(A_1, C) \leq (1+\varepsilon)f(S_1, C)$$

$$(1-\varepsilon)f(S_2, C) \leq f(A_2, C) \leq (1+\varepsilon)f(S_2, C)$$

相加得

$$(1-\varepsilon)f(S_1 \cup S_2, C) \leq f(A_1 \cup A_2, C) \leq (1+\varepsilon)f(S_1 \cup S_2, C)$$

即 $S_1 \cup S_2$ 是 $A_1 \cup A_2$ 的 (K, ε) -coreset

Problem 7

$$1. E(\hat{n}) = E\left(\frac{(1+\alpha)^{X_n} - 1}{\alpha}\right)$$

$$E[(1+\alpha)^{X_{n+1}}] = \sum_{i=1}^{\infty} \Pr[X_n = i] \cdot E[(1+\alpha)^{X_{n+1}} | X_n = i]$$

$$= \sum_{i=1}^{\infty} \Pr[X_n = i] \cdot \left[(1+\alpha)^{i+1} \cdot \frac{1}{(1+\alpha)^i} + (1+\alpha)^i \cdot \left(1 - \frac{1}{(1+\alpha)^i}\right) \right]$$

$$\begin{aligned}
&= \sum_{i=1}^{\infty} \Pr[X_n=i] \left[1+\alpha + \frac{(1+\alpha)^i}{-1} \right] \\
&= \sum_{i=1}^{\infty} \Pr[X_n=i] \cdot \alpha + \sum_{i=1}^{\infty} \Pr[X_n=i] (1+\alpha)^i \\
&= \alpha + E[(1+\alpha)^{X_n}]
\end{aligned}$$

$$E[(1+\alpha)^{X_0}] = 1$$

$$\therefore E[(1+\alpha)^{X_n}] = \alpha n + 1$$

$$\Rightarrow E[\hat{n}] = \frac{E[(1+\alpha)^{X_n}] - 1}{\alpha} = n$$

$$\text{Var}(\hat{n}) = E[\hat{n}^2] - E[\hat{n}]^2$$

$$E[\hat{n}]^2 = E\left[\frac{(1+\alpha)^{2X_n} - 2(1+\alpha)^{X_n} + 1}{\alpha^2} \right]$$

$$\text{if } E[(1+\alpha)^{2X_n}]$$

$$\begin{aligned}
E[(1+\alpha)^{2X_{n+1}}] &= \sum_{i=0}^{\infty} \Pr(X_n=i) E[(1+\alpha)^{2X_{n+1}} | X_n=i] \\
&= \sum_{i=0}^{\infty} \Pr(X_n=i) \left[(1+\alpha)^{2(i+1)} \cdot \frac{1}{(1+\alpha)^i} \right. \\
&\quad \left. + (1+\alpha)^{2i} \cdot \left(1 - \frac{1}{(1+\alpha)^i} \right) \right] \\
&= \sum_{i=0}^{\infty} \Pr(X_n=i) \left[(1+\alpha)^{i+2} + (1+\alpha)^{2i} \right]
\end{aligned}$$

$$\begin{aligned}
 & - (1+\alpha)^i \Big] \\
 & = (\alpha^2 + 2\alpha) \sum_{i=0}^{\infty} \Pr(x_n = i) (1+\alpha)^i \\
 & \quad + \sum_{i=0}^{\infty} \Pr(x_n = i) (1+\alpha)^{2i} \\
 & = (\alpha^2 + 2\alpha) E[(1+\alpha)^{x_n}] + E[(1+\alpha)^{2x_n}]
 \end{aligned}$$

递推有

$$E(\hat{n})^2 = \frac{\frac{n^2}{2} \alpha^3 + \frac{n}{2} \alpha^3 + \alpha^2 n^2 + \alpha^2 n - 2\alpha n}{\alpha^2}$$

$$\begin{aligned}
 \text{Var}(\hat{n}) &= E(\hat{n}^2) - [E(\hat{n})]^2 \\
 &= \frac{\alpha}{2} (n^2 - n)
 \end{aligned}$$

2. 即 Morris + 算法

- ① 独立运行 S 个 Morris 算法, 获得 $\hat{n}_1 \dots \hat{n}_S$
- ② 在时刻 n 查询时输出 $\hat{n} = \frac{1}{S} \sum_{i=1}^S \hat{n}_i$

$$E(\hat{n}) = n \quad (\text{显然})$$

$$\begin{aligned}
 \text{Var}(\hat{n}) &= \frac{1}{S^2} \sum_{i=1}^S \text{Var}(\hat{n}_i) = \frac{1}{S^2} \alpha \left(\frac{n^2}{2} - n \right) \\
 &= \frac{\alpha n^2}{2S}
 \end{aligned}$$

$$\Pr[|\hat{n} - n| > \varepsilon n] = \Pr[|\hat{n} - E(\hat{n})| > \varepsilon n] \leq \frac{\text{Var}(\hat{n})}{\varepsilon^2 n^2}$$

$$= \frac{\alpha}{2S\varepsilon^2}$$

$$\sqrt{\frac{\alpha}{2S\varepsilon^2}} = \delta \quad \text{有 } S = \frac{\alpha}{2\delta^2\varepsilon^2}$$

$$S = \frac{\alpha}{2\delta^2\varepsilon^2} \text{ 时满足条件}$$

思考题1: 提高采样效率, 增大步长, 使其与时间的相关性减弱

思考题2: 使用 K-means++ 算法, 设定 K 为核心集且核心大小 10 倍, 生成 Coreset